

CSE6214 Software Engineering Fundamentals

Tutorial 11

Part A: Discussion

Topic (Lecture 11): Software Testing Strategies

1. Describe the difference between verification and validation.
2. Why is a highly coupled module difficult to unit test?
3. Why is regression testing an important part of any integration testing procedure?

Part B: Tutorial Activity

Topic: Software Testing Strategies

Match the following classes and approaches of testing with its descriptions. You should select the best answer from Table 1.

Table 1 – List of Tests Classes and Approaches.

Deployment testing	Stress testing	Black-box testing
Performance testing	Beta test	Recovery Testing
White-box testing	Unit testing	Bottom-up Integration Testing
Regression Testing	Alpha testing	Top-down Integration Testing
Security testing	Smoke testing	Component testing

1. _____ focuses verification effort on the smallest unit of software design, the software component or module.
2. _____ begins construction and testing with atomic modules which refers to the components at the lowest levels in the program.
3. _____ is an incremental approach to construction of the software architecture.
4. _____ attempts to verify that protection mechanisms built into a system will protect it from improper penetration.
5. _____ is the re-execution of some subset of tests that have been conducted to ensure that changes have not propagated unintended side effects.
6. _____ provides a number of benefits when it is applied on complex, time-critical software projects.
7. _____ is conducted in a controlled environment. It is conducted at the developer's site by a representative group of end user.
8. _____ is a system test that forces the software to fail in a variety of ways and verifies that recovery is properly performed.

9. _____ executes a system in a manner that demands resources in abnormal quantity, frequency or volume.
10. _____ is also called configuration testing. It examines all installation procedures and specialized installation software that will be used by customers, and all the documentation that will be used to introduce the software to end-users.
11. _____ is designed to test the run-time performance of software within the context of an integrated system.
12. _____ is an integrating test approach that is designed as a pacing mechanism for time-critical projects, allowing the software team to assess the project in a frequent basis.
13. _____ is a testing that is conducted at one or more end-user sites.
14. _____ or also called as glass-box testing is a test-case design philosophy that uses the control structure described as part of component-level design to derive test cases.
15. _____ examines some fundamental aspects of a system with little regard for the internal logical structure of the software.

Part C: Project

Task: Software integration & integration testing

1. Based on the data dictionary, create a database with all the defined tables for the software. For example, you can define the classes in 'models.py' in the Django framework and execute 'python manage.py makemigrations' and 'python manage.py migrate' to update the database. (Refer to <https://docs.djangoproject.com/en/4.1/topics/db/models/> for more details)
2. Determine a strategy to integrate all the subsystems e.g. top-down or bottom-up. Explain the strategy to the tutor.
3. Complete the integration of the subsystems based on the integration strategy.
4. Perform the necessary integration testing as the software integration is done.